Nishka Mathew

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SUMMARY

UX Researcher and Data Analyst with a published paper in ACM COMPASS 2025 and a dual degree in Computer Science and Psychology from UNC Charlotte (GPA: 3.83). Combines mixed-methods research, qualitative synthesis, and quantitative analysis to translate user behavior into actionable insights for non-technical stakeholders. Experienced applying human-centered principles to AI-driven systems, with a focus on human-AI interaction and user trust.

EDUCATION

The University of North Carolina at Charlotte ***GPA: 3.83***

Bachelor of Arts in Computer Science and Bachelor of Science in Psychology

- **Relevant Coursework**: Research Methodology II, Data Structures & Algorithms, Data Mining, Human-Centered AI, Cognitive Science, iOS Development, Design Prototyping, UX Methods, Elements of Statistics II

Honors: Chancellor's List (Fall 2022 - Fall 2023, Fall 2024, Fall 2025) & Dean's List (Spring 2024, Spring 2025)

WORK EXPERIENCE

UX Research Analyst Intern at Security, Privacy and User Experience (SPEX) Lab **May 2024 – Dec 2025**

- Coauthored a peer-reviewed research paper published to **ACM COMPASS 2025** examining how **40** first-generation international students across **14** countries manage privacy across social media platforms throughout **3** migration stages
- Conducted **25+** semi-structured interviews, synthesizing qualitative data through qualitative coding to identify **5** key themes shaping privacy practices; analyzed quantitative data using **IBM SPSS Statistics** to create data visualizations
- Prototyped a privacy-centered AI chatbot via **OpenAI API** and diagnosed usability friction through heuristic evaluation

Teacher's Assistant at the UNC Charlotte **Jan 2023 – May 2024**

Spring 2023: College Algebra, **Fall 2023**: Research Methodology I**, Spring 2024**: Research Methodology II

- Facilitated workshops and tutoring sessions for **200+** students across **3** courses, improving performance by **12-15%** through iterative refinement of instructional materials based on student feedback
- Guided **30+** student research projects from initial proposal through mock **IRB** submission, ensuring ethical compliance

PROJECTS

AI-Powered Microalgae Biomass Predictor for Ecoclosure Biochromic Windows

- Developed predictive **ML** models using a Kaggle sample dataset (9,784 data points) and real-world sensor data achieving **R^2=0.911** using **Python**, **Scikit-learn**, and **Pandas** to forecast multivariate biomass growth
- Designed an interactive visualization dashboard in **Figma** and **React** to overlay historical trends with **14-day** predictive projections, translating complex model outputs into an interpretable **human-centered AI** experience increasing trust

MvR: Movie & TV Review Application 3 Phase UX Research Study

- Led a **3**-phase UX research and design study with **15** usability / concept testing interviews, co-design workshops with **10** participants, and iterative Figma prototyping to find **5** user need themes across trust, discovery, and personalization
- Delivered empathy maps, affinity diagrams, personas, user journey maps, wireframes, insight summaries, and low-to-high fidelity prototypes; applied heuristic evaluation at each iteration and practiced design QA throughout
- Through iterative redesigns, final prototype (78) increased **SUS** score by **9** points over baseline competitive apps (67)

Pearl Hacks 2025 – 1st Place Sponsored by Qorvo

- Built and shipped a responsive mentorship platform in **24** hours with a **Flask/Python** back-end, **SQLAlchemy** database, **NLTK** keyword matching for mentorship filtering, and live **D3.js** network visualizations
- Owned information architecture, interaction flows, and design system across the entire application; practiced design QA to maintain fidelity between design and shipped interface

SKILLS

Python, SQL, R, Pandas, Scikit-learn, Jupyter Notebook, IBM SPSS Statistics, Dovetail, D3.js, Qualtrics, Tableau, PowerBi, JavaScript, CITI Human Subjects Research Certification, Postman, Miro, Claude, Copilot, HTML, Figma, GitHub, React